SPLEEN

Anatomy

- Weight **75–250 g**, ~10×7×3 cm, **left hypochondrium** between gastric fundus + left hemidiaphragm, long axis along **10th rib**.
- **Gastrosplenic ligament** (anterior to hilum) → greater curvature stomach (contains short gastric + left gastroepiploic vessels).
- **Splenorenal ligament** (posterior to hilum) → left kidney; contains **splenic vessels + tail of pancreas**.
- **Splenic artery** from coeliac axis, runs along upper border of pancreas. **Splenic vein** + SMV → **portal vein**.

Functions (5)

1. **Immune**: ~70% T + 10–15% B lymphocytes; major site of **IgM**; makes opsonins (properdin, tuftsin). Loss → must vaccinate vs capsulated organisms.
2. **Filter**: macrophages remove effete RBCs/platelets, bacteria (esp. pneumococci); recycles iron.
3. **Pitting**: removes RBC inclusions (**Howell–Jolly, Heinz bodies, target cells**). Lost in sickle cell **autosplenectomy**.
4. **Reservoir**: ~8% RBC mass; massive enlargement → pancytopenia (correctable by splenectomy).
5. **Cytopoiesis**: fetal haemopoiesis; reactivates in myeloproliferative/haemolytic disease.

Investigation

- Haemolytic anaemia: FBC, reticulocytes. Portal hypertension: liver signs, varices, pancytopenia.
- **US** (size/cyst), **CT with contrast** (characterise pathology — main), MRI (T2 for cysts). Plain film: calcification (aneurysm, old infarct, cyst, hydatid, TB).

Congenital abnormalities

- **Agenesis** (rare; 5% of congenital heart disease). **Polysplenia** (failed fusion).
- **Splenunculi (accessory spleens)**: 10–30% population; 50% near hilum; **must remove at splenectomy or disease persists**.
- Hamartomas (rare). **Non-parasitic cysts**: true (epithelial lining — dermoid, epidermoid) vs pseudocyst (no lining — common **after pancreatitis**).

Splenic artery aneurysm

- 0.04–1% at autopsy; **twice as common in women**; main trunk; usually single.
- Cause: intra-abdominal sepsis, pancreatic necrosis, arteriosclerosis (elderly).
- **Asymptomatic until rupture** (mimics splenic rupture). **¼ of ruptures in pregnant women (3rd trimester/labour)**.
- Treatment: **embolisation / endovascular stenting** (now preferred) or splenectomy; calcified in elderly → observe.

Splenic infarction

- Massive spleen (myeloproliferative, portal HTN), splenic vein thrombosis, sickle cell.
- Asymptomatic or **LUQ + left shoulder tip pain**. Conservative; splenectomy only if septic infarct → abscess.

Splenic rupture (trauma)

- **Spleen = most commonly injured intra-abdominal organ** (then liver). Suspect in blunt LUQ trauma. Also iatrogenic.
- **CT = most important** (haemodynamically stable); graded I–V.
- **Management**: vigorous resuscitation. **Splenectomy for severe grades** (bleeding control > salvage). **Angiography + embolisation** may avoid splenectomy but **must NOT delay laparotomy if unstable**.

Splenomegaly & hypersplenism

- Spleen enlarges **threefold before palpable**.
- **Causes**: infective (malaria, EBV, typhoid, TB, kala-azar, schistosomiasis), blood (leukaemia, thalassemia, spherocytosis, sickle, AIHA, polycythaemia), metabolic (Gaucher's, amyloid), circulatory (**portal hypertension**, infarct), collagen (Felty's, Still's), cysts, neoplastic (lymphoma, myelofibrosis).
- **Hypersplenism**: splenomegaly + cytopenia (anaemia/leukopenia/thrombocytopenia) + marrow hyperplasia + **improves after splenectomy**.

Neoplasms

- **Haemangioma = most common benign** (rarely → haemangiosarcoma).
- **Lymphoma = most common cause of neoplastic enlargement**; metastases rare.
- Myelofibrosis: marrow proliferation → splenomegaly; splenectomy reduces transfusion need.

Splenectomy

- **Indications**: trauma (accidental/iatrogenic); oncological (en bloc); **haematological (spherocytosis, ITP, hypersplenism)**; portal hypertension (variceal surgery).
- **Preop**: correct coagulation (FFP, platelets, cryo); antibiotic prophylaxis; **vaccinate — pneumococcus + meningococcus C (every 5 yrs), Hib (every 10 yrs)**.
- **Complications**: haemorrhage (slipped ligature), **left basal atelectasis + pleural effusion**, pancreatic tail injury (pancreatitis/fistula/abscess), gastric dilatation, **thrombocytosis**, **OPSI (overwhelming post-splenectomy infection)**.

High-yield one-liners

- **Spleen = most commonly injured abdominal organ; CT for stable, laparotomy if unstable.**
- **Pitting failure → Howell–Jolly + Heinz bodies + target cells** (post-splenectomy/autosplenectomy).
- **Splenunculi (10–30%) must be removed** or disease recurs.
- **Splenic artery aneurysm: women, ruptures in pregnancy; embolisation.**
- **Hypersplenism = splenomegaly + cytopenia + improves post-splenectomy.**
- **Lymphoma = commonest neoplastic splenomegaly; haemangioma = commonest benign tumour.**
- **Post-splenectomy: vaccinate (pneumococcus/meningococcus/Hib) + OPSI risk.**